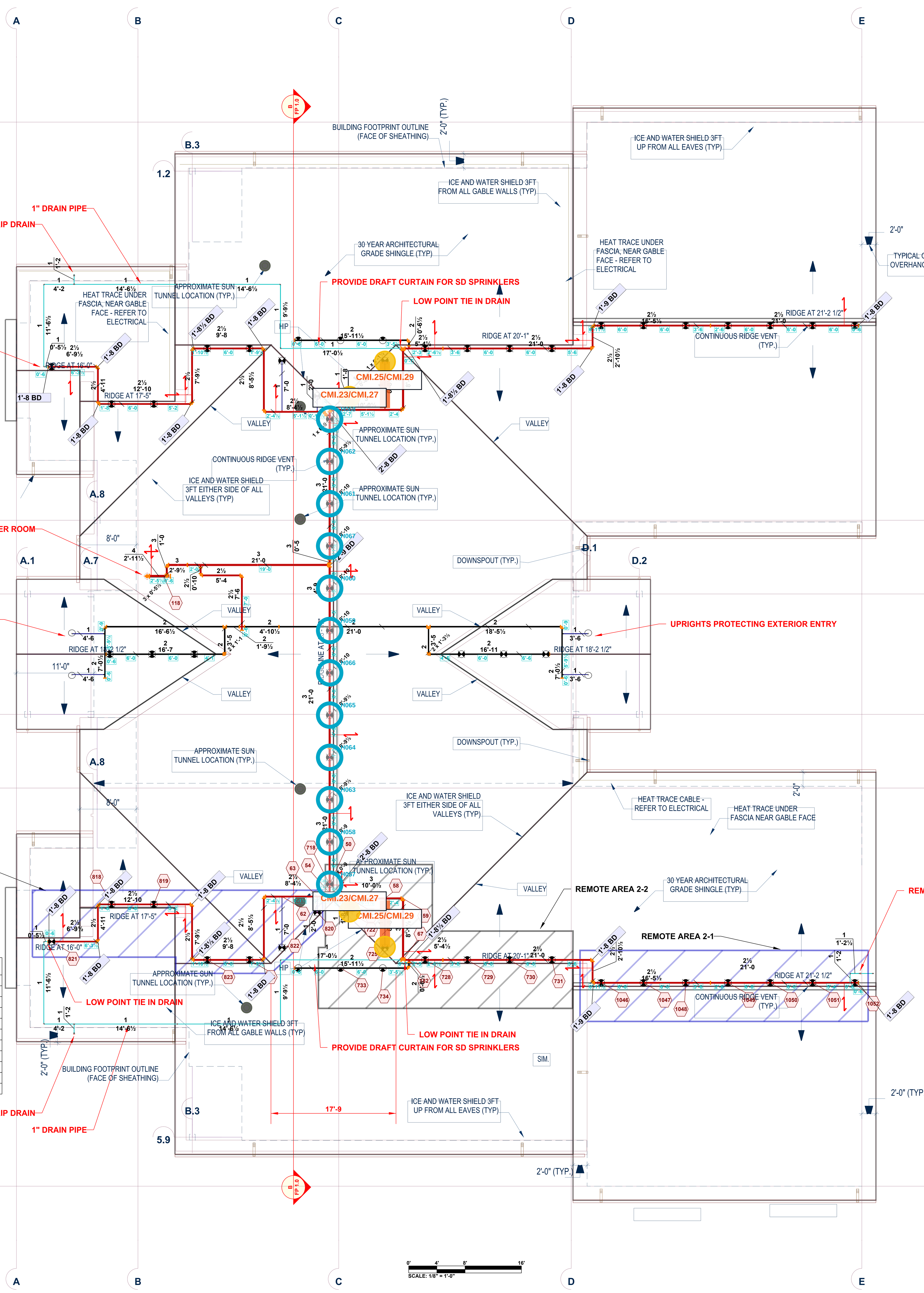
NEW HOPE CMI.23-CMI.42 - ACTUAL FP2.0 TOP VIEW

4 approved horizontal arm-over routes | 12 exact direct carrier/head endpoints | equal piece identities remain intentionally unresolved

Sprinkler Legend											
Symbol	Manufacturer	SIN	Model	Quantity	K-Factor	Type	Size	Response	Finish	Temperature	Note
	Tyco	TY4180	BB1	58	8	Upright	¾	Quick	Brass	200°F	
	Tyco	TY3183	SD1	6	5.6	Upright	½	Quick	Brass	212°F	
	Tyco	TY3131	TY-FRB	4	5.6	Upright	½	Quick	Natural Brass	200°F	
				Total = 68							



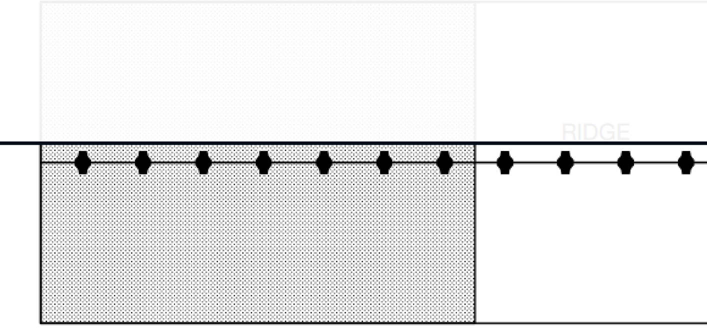
Hydraulic Information	
Remote Area 2-3	
OCCUPANCY CLASSIFICATION	Light Hazard
DENSITY (gpm/ft²)	0.10 for 414ft² (Actual 414ft²)
TOTAL HOSE STREAMS	100.0
TOTAL HEADS FLOWING	7
K-FACTOR	8
TOTAL WATER REQUIRED	379.5
TOTAL PRESSURE REQUIRED	67.9
BASE of RISER (gpm)	279.5
BASE of RISER (psi)	53.7
SAFETY MARGIN (psi)	+11.1 (14.1%)

Hydraulic Information	
Remote Area 2-1	
OCCUPANCY CLASSIFICATION	Light Hazard
DENSITY (gpm/ft²)	0.10 for 422ft² (Actual 412ft²)
TOTAL HOSE STREAMS	100.0
TOTAL HEADS FLOWING	7
K-FACTOR	8
TOTAL WATER REQUIRED	371.5
TOTAL PRESSURE REQUIRED	73.6
BASE of RISER (gpm)	271.5
BASE of RISER (psi)	59.9
SAFETY MARGIN (psi)	+5.4 (6.8%)

Hydraulic Information	
Remote Area 2-2	
OCCUPANCY CLASSIFICATION	Light Hazard
DENSITY (gpm/ft²)	0.10 for 537ft² (Actual 537ft²)
TOTAL HOSE STREAMS	100.0
TOTAL HEADS FLOWING	9
K-FACTOR	8
TOTAL WATER REQUIRED	438.4
TOTAL PRESSURE REQUIRED	69.6
BASE of RISER (gpm)	338.4
BASE of RISER (psi)	53.7
SAFETY MARGIN (psi)	+9.4 (11.9%)

FIGURE 20-A-1. BB SPRINKLERS

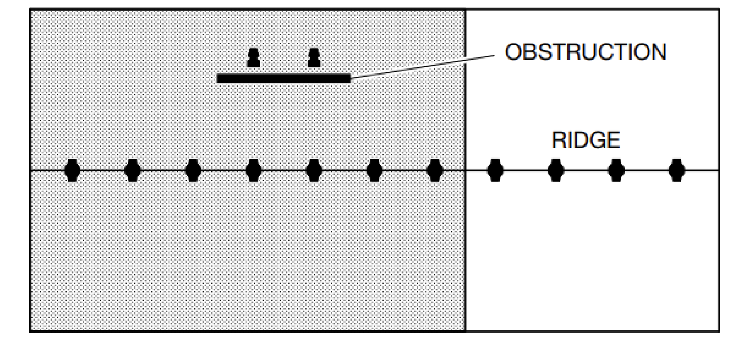
- Wet Systems: Calculate the most demanding five sprinklers.
- Dry Systems: Calculate the most demanding seven sprinklers. See the adjacent figure.



DRY SYSTEM SHOWN

FIGURE 20-A-3. BB AND SD SPRINKLERS

- Wet Systems: Calculate the most demanding five BB Sprinklers plus two SD Sprinklers.
- Dry Systems: Calculate the most demanding seven BB Sprinklers plus up to two SD Sprinklers. See the adjacent figure.



DRY SYSTEM SHOWN

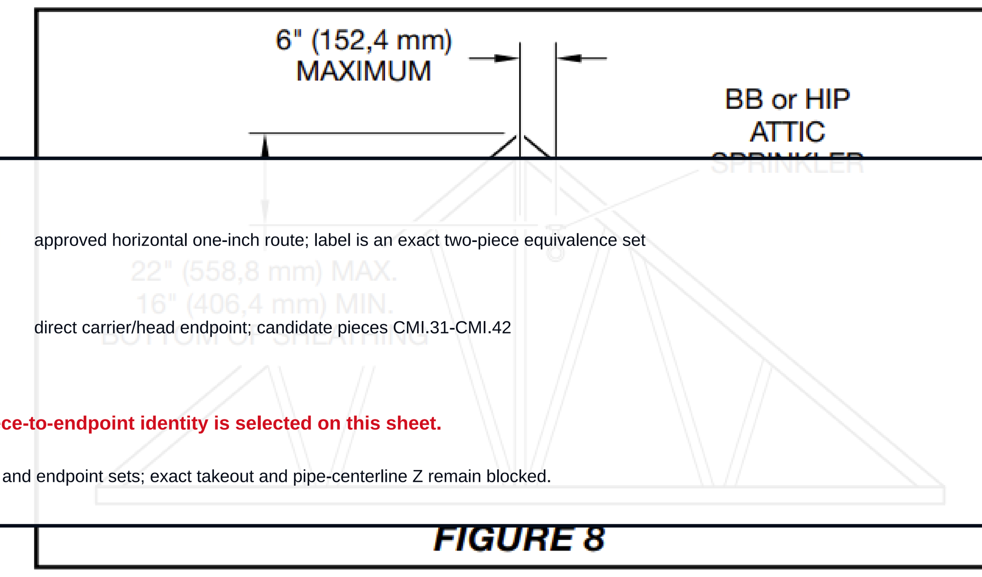


FIGURE 8

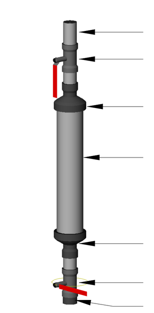
approved horizontal one-inch route; label is an exact two-piece equivalence set

direct carrier/head endpoint; candidate pieces CMI.31-CMI.42

NO fabricated piece-to-endpoint identity is selected on this sheet.

Top view proves XY and endpoint sets; exact takeout and pipe-centerline Z remain blocked.

DRUM DRIP DETAIL



DRY PIPE NOTE:
SLOPE BRANCH LINES ½" EVERY 10'-0"
SLOPE CROSS MAINS ¼" EVERY 10'-0"

NOTE: ADDITIONAL SPRINKLER HEADS OVER AND ABOVE THOSE SPECIFICALLY NOTED ON THESE DRAWINGS MAYBE REQUIRED DEPENDING ON THE REQUIRED ROUTING AROUND TRUSSES, HVAC DUCTWORK, HOODS AND OTHER SIMILAR OBSTRUCTIONS. FIELD VERIFY EXISTING CONDITIONS DURING CONSTRUCTION AND PROVIDE ANY AND ALL ADDITIONAL HEADS AS REQUIRED FOR FULL COVERAGE AND A COMPLETE, CODE COMPLIANT SYSTEM.

FIRE SPRINKLER LAYOUT - ATTIC
1/8" = 1 Foot

HFP DESIGN
HFP DESIGN, LLC
Phone: 602-628-0457
Email: Cole.howard@hfpdesign.com

COLE HOWARD, CET
NICET LEVEL III
FIRE PROTECTION ENGINEERING TECHNOLOGY
CERTIFICATE #151494 (EXP 12/1/2025)

Cole Howard 03/03/25
NICET III # 151494
Exp. 12/1/25

HaloFire
4811 E. Julep St., Ste 129
Mesa, Arizona 85205
Phone (480) 325-2280
www.HaloFireUS.com

SPRINKLER INSTALLER MAY SUBSTITUTE A SPRINKLER FROM A DIFFERENT MANUFACTURER AS LONG AS ALL CRITERIA (K-FACTOR, RESPONSE, THREAD, TEMP, COLOR, ETC) ARE THE SAME.

REVISIONS			
NO.	DATE	DESCRIPTION	BY
	3/3/2025	FIRE MARSHAL REVIEW COMMENTS	HFPD

DRAWN BY: HFPD
DATE: 1/29/2025
PLOT DATE: 03/03/25

A New Building For:
New Hope Crisis Center
650 E 700 S
BRIGHAM CITY, UTAH 84302
FIRE SPRINKLER SYSTEM - ATTIC

CONTRACT NUMBER
24-052

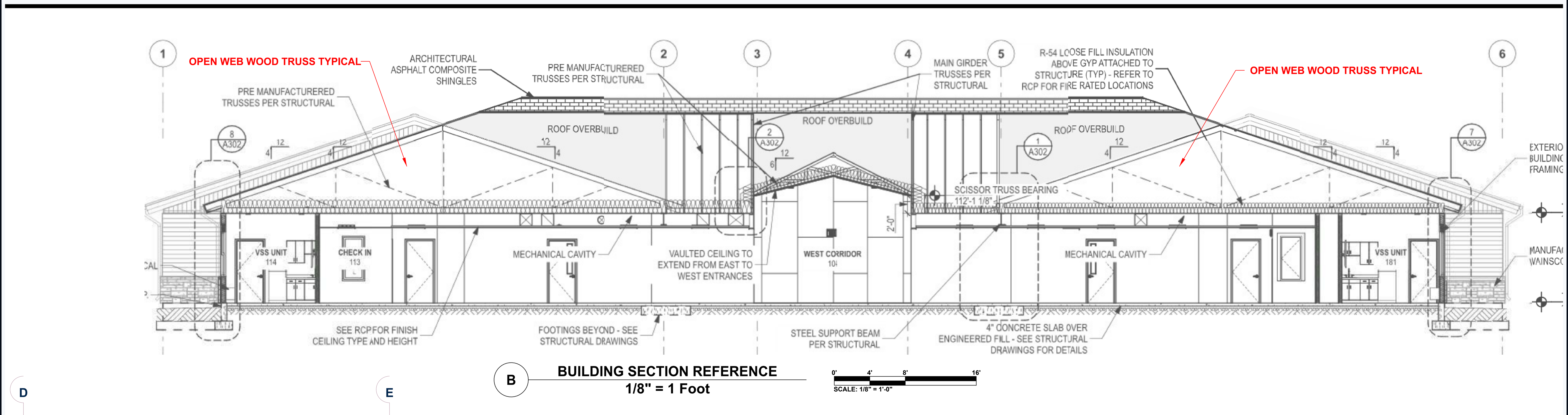
DRAWING FILE NAME
NHCC_FIRE

AUTHORITY
BRIGHAM CITY F.D.

SHEET NUMBER
FP 2.0

CMI THREADED TERMINALS - ELEVATION EVIDENCE BOUNDARY

Actual FP1.0 building section + actual listing page 42 | schematic classes only; no pipe is placed into the section without exact Z

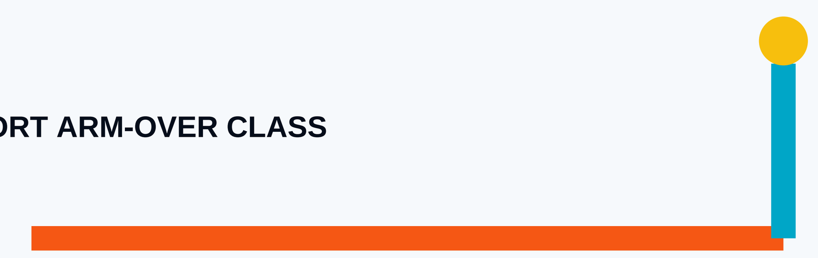


ACTUAL FIELD FP1.0 BUILDING SECTION - PIPE OVERLAY WITHHELD: per-piece Z is not published

.10	1 - Pipe, Schedule 40	0'-4½	T - T	1 x ¾ Threaded Reducer, DI
Line: CMI- Quantity: 1				
.23	1 - Pipe, Schedule 40	1'-8	T - T	1 Threaded, 90° Elbow, DI
.24	1 - Pipe, Schedule 40	0'-8½	T - T	1 x ¾ Threaded Reducer, DI
.25	1 - Pipe, Schedule 40	6'-8½	T - T	1 Threaded, 90° Elbow, DI
.26	1 - Pipe, Schedule 40	0'-1½	T - T	1 x ¾ Threaded Reducer, DI
.27	1 - Pipe, Schedule 40	1'-8	T - T	1 Threaded, 90° Elbow, DI
.28	1 - Pipe, Schedule 40	0'-8½	T - T	1 x ¾ Threaded Reducer, DI
.29	1 - Pipe, Schedule 40	6'-8½	T - T	1 Threaded, 90° Elbow, DI
.30	1 - Pipe, Schedule 40	0'-1½	T - T	1 x ¾ Threaded Reducer, DI
.31	1 - Pipe, Schedule 40	0'-10	T - T	1 x ¾ Threaded Reducer, DI
.32	1 - Pipe, Schedule 40	0'-10	T - T	1 x ¾ Threaded Reducer, DI
.33	1 - Pipe, Schedule 40	0'-10	T - T	1 x ¾ Threaded Reducer, DI
.34	1 - Pipe, Schedule 40	0'-10	T - T	1 x ¾ Threaded Reducer, DI
.35	1 - Pipe, Schedule 40	0'-10	T - T	1 x ¾ Threaded Reducer, DI
.36	1 - Pipe, Schedule 40	0'-9½	T - T	1 x ¾ Threaded Reducer, DI
.37	1 - Pipe, Schedule 40	0'-9½	T - T	1 x ¾ Threaded Reducer, DI
.38	1 - Pipe, Schedule 40	0'-9	T - T	1 x ¾ Threaded Reducer, DI
.39	1 - Pipe, Schedule 40	0'-9½	T - T	1 x ¾ Threaded Reducer, DI
.40	1 - Pipe, Schedule 40	0'-9½	T - T	1 x ¾ Threaded Reducer, DI
.41	1 - Pipe, Schedule 40	0'-9	T - T	1 x ¾ Threaded Reducer, DI
.42	1 - Pipe, Schedule 40	0'-9	T - T	1 x ¾ Threaded Reducer, DI
Line: CMJ- Quantity: 1				
.08	1 - Pipe, Schedule 40	7'-0	T - T	1 x ¾ Threaded, 90º Reducing Elbow, DI
.09	1 - Pipe, Schedule 40	2'-0	T - T	1 Threaded, 90° Elbow, DI
.10	1 - Pipe, Schedule 40	0'-4½	T - T	1 x ¾ Threaded Reducer, DI

ACTUAL LISTING PAGE 42 - exact cut lengths, T-T ends, and fitting families

SHORT ARM-OVER CLASS



LONG ARM-OVER CLASS

DIRECT VERTICAL NIPPLE CLASSES - 12 exact carrier/head endpoints

CMI.31-CMI.35

5 pieces x 10 in

CMI.36/.37/.39/.40

4 pieces x 9.5 in

CMI.38/.41/.42

3 pieces x 9 in

FAIL-CLOSED IDENTITY BOUNDARY

PROVED

- 20 exact CMI.23-CMI.42 piece identities
- 20 exact cut lengths and fitting families
- 4 approved horizontal FP2.0 routes
- 12 exact direct carrier/head endpoints
- 7 deterministic equivalence classes

NOT PROVED

- exact piece-to-route identity inside equal classes
- exact direct nipple-to-head identity
- connection takeout
- complete vertical offsets and whole-system Z

7,664,025,600 exact identity assignments remain.

No assignment is fabricated by this proof.